Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/01/2025
PI Name:	Wu, Martin (Chi Heng)		
Project Title:	Improvement of Shelf Life for Space Food Through Hurdle Approach		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food**:Risk of Performance Decrement and Crew Illness Due to an Inadequate Food System		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2019	**End Date:**	03/31/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Start date revised to 12/01/2019 from 10/01/2019 per discussions with PI and HRP (Ed., 8/2/21)		
Key Personnel Changes/Previous PI:	Summary as per PI (September 2023 Update): Research and reporting responsibilities were transferred from Takiyah Sirmons (October 2019) to Corunda T. Pruitt (March 2022) and then to Martin (Chi Heng) Wu (May 2023). May 2023--Martin (Chi Heng) Wu, Ph.D. has taken over the project as PI. Since Dr. Wu is now PI, so his name has been removed from the list of Co-Investigators (Ed., 5/31/23). Summary of Reporting (Ed., September 2023): Research and reporting responsibilities were transferred from Dr. Maya Cooper (April 2019) to Dr. Takiyah Sirmons (October 2019) to Dr. Martin Wu (May 2023). Please visit the NASA Task Book for reports from specific fiscal years: https:// (Wu) https:// (Sirmons) https:// (Cooper)		
COI Name (Institution):	Froio-Blumsack, Danielle M.S. (U.S. Army Natick Soldier RD&E Center) Douglas, Grace Ph.D. (NASA Johnson Space Center) Young, Millenia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study investigated the potential to extend the shelf life of foods treated with multiple stabilization technologies in combination with lower temperature storages (a combination of hurdle approaches) beyond its original specification, mostly within 1 to 3 years, to 5 years. There were 33 foods produced with different food stabilization technologies, such as sterilization (either RETORT, MATS, or IRR), packages (various levels of vacuum) or ingredients (either fresh or frozen), and stored at various temperature (21, 4, -20 or -80 oC) throughout the shelf life. At designated time points, each sample is rated on appearance, color, aroma, flavor, texture, and overall acceptability on a 9-point hedonic scale where 1 corresponds to “dislike extremely” and 9 corresponds to “like extremely”. Selected samples are evaluated for chosen nutrients following the Official Methods of Analysis of AOAC International for each micronutrient. Data to date indicates it is essential to store foods, such as apricot cobbler, dried apricots, peaches, strawberries, carrot coins, Italian vegetables, tomato basil soup, beef steak, grilled chicken, broccoli, green beans, macaroni noodles, potatoes, spinach, quinoa, cheese sauce, ketchup, kraft mayo, mustard, soy sauce and wheat flat bread (21 out of 33) at lower temperature to extend its shelf life beyond 3 years because the food decreased remarkably in either overall acceptability, nutrients, or both. The other foods might require data from future time points (60, 84 months or both) to confirm trends and requirements since the sensory attributes and nutrients remained either stable or similar among the various hurdle treatments over shelf life.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Advanced Food Technology Project (AFT) aims to develop the requirements for a food system that will provide the crew with a safe, nutritious, and acceptable food system while remaining within the constraints of available vehicle resources such as mass, volume, and crew preparation time on exploration missions (Cooper et al,. 2011). The food system on the International Space Station is composed of prepackaged thermally stabilized foods, irradiated foods, freeze-dried foods, intermediate moisture or bite sized foods, and powdered drinks, all stored at ambient conditions (currently 21ºC) due to resource constraints. In the case of long exploratory missions, this same prepackaged food system could continue as the primary food system. The longer mission requires the food system to sustain the crew for three to five years without replenishment. However, many of the space menu items do not maintain acceptability or nutritive value for a five-year period using current stabilization strategies (Catauro and Perchonok 2012, Barrett and Cardello 2012). Sensory acceptability is critical to ensure crew maintain adequate levels of consumption. Inadequate food acceptability has been linked to decreased food consumption, which may affect crew nutrition and psychosocial health and limit the crew’s ability to complete mission-critical tasks (Friedl and Hoyt 1997). The adequacy of the food system becomes increasingly important in the harsh environments of isolation and confinement, where other comforts and familiarities are unavailable (Stuster 2000). The processed and prepackaged space food system is the main source of crew nutrition, and hence is central to astronaut health and performance. Unfortunately, space food quality and nutrition degrade to unacceptable levels in 1 to 3 years with current food stabilization technologies. Future exploration missions will require a food system that remains safe, acceptable, and nutritious through 5 years of storage within vehicle resource constraints. This study assesses the potential of combining stabilization technologies (alternative storage temperatures, processing, formulation, ingredient source, packaging, food components and preparation procedures) to mitigate degradation of quality and nutritional content in space food. This update focuses on the aspects of sensory and nutritional stability of 33 foods and condiments through their current shelf-life timepoints, which varies between 3 and 5 years for each food.
Task Progress:	This project aimed to investigate over 33 foods with various combinations of different stabilization hurdle approaches, including processing, packaging, and storage temperatures. This report will focus on the sensory and nutritional analysis throughout the shelf life. In addition, some of the listed foods will be evaluated for 7 years, with testing continuing through 2027. Interpretation of this information is ongoing and will be available in a future report.
Bibliography Type:	Description: (Last Updated:)